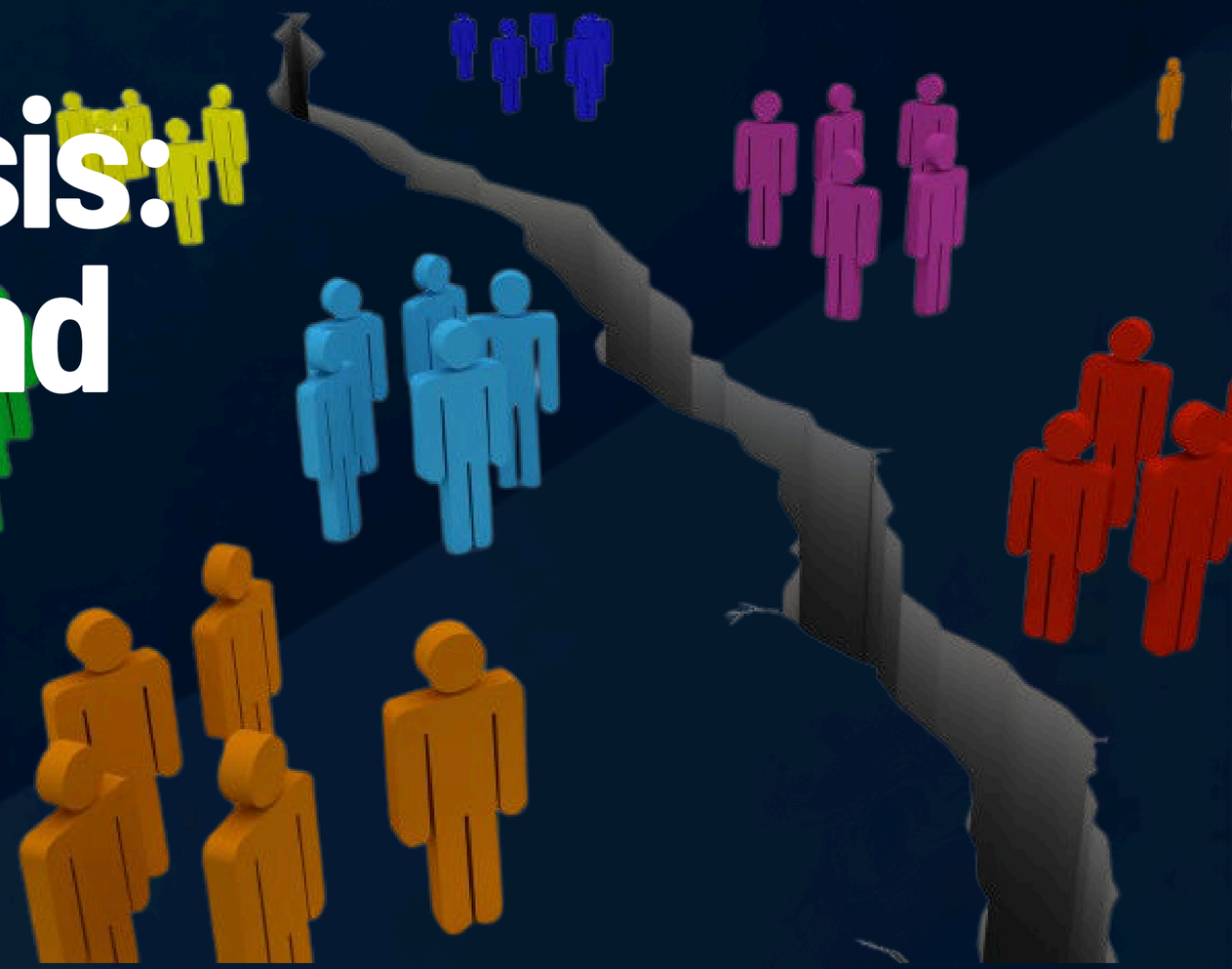
Burmese Online Polarization Analysis: Detection, Type, and Manifestation



Course / Assignment : AIEF - 2 , Assignment 7

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WHAT IS POLARIZATION?

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Fact that people or opinions being divided into two opposing groups



Main components:

- Task 1 - Identifying Polarization
- Task 2 - Classifying Polarization (Types)
- Task 3 - Identifying Polarization Expression



Our project focuses on all tasks by Burmese language



We implemented both:

- Traditional ML models
- Neural Network-based approaches

PROJECT OBJECTIVES

Task 1 — Polarization Detection

- Determine whether a text is polarized or non-polarized.
- Binary classification.

Task 2 — Polarization Type Classification

- Identify the type of polarization expressed.
- Five target categories.

Task 3 — Polarization Manifestation Identification

- Identify how polarization is expressed in text.
- Six target categories.
- Dataset: myPolar raw data prepared for Assignment 2.

DATASET PREPARATION

Stage 1: Annotation Master Dataset

- **Original Burmese text**
- **Source or resource link**
- **Annotator name**
- **Annotation labels**
- **Record ID and other relevant metadata**

Stage 2: Model-Ready Dataset

- **Cleaned text**
- **Task-specific target labels**
- **Consistent label formats**
- **No unnecessary metadata used as model features**
- **Final dataset: 10,350**

DATASET PREPARATION

KEY CHALLENGES

- Inconsistent formats across annotators
- Missing or invalid labels
- Duplicate records
- Malformed rows and text-label misalignment

EXPERIMENTAL WORKFLOW



TASK 1 — POLARIZATION DETECTION

Question: Is the given Burmese text polarized?

Feature Extraction

- Count Vectorization
- TF-IDF
- Word unigrams and bigrams

Data Split: 70% training, 15% validation, 15% testing.

Models Evaluated

- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- SGD Classifier
- Linear SVM
- Bidirectional LSTM (BiLSTM)
- Stacked BiLSTM

TASK 1 — EXPERIMENTAL RESULTS

System	Macro F1
Random Forest + Bigram Count	0.7103
Tuned Linear SVM	0.6751
BiLSTM	0.7091
Stacked BiLSTM	0.7188

Best Reported Model

Stacked BiLSTM

TASK 2 — POLARIZATION TYPE CLASSIFICATION

Question: What type of polarization is expressed in text?

Classification Type: Multi-label Classification

Five Target Categories

- **Political**
- **Racial / Ethnic**
- **Religious**
- **Gender / Sexual**
- **Other**

Models Evaluated

- **BiLSTM**
- **Word TF-IDF + Linear SVM**
- **Character TF-IDF + Linear SVM**
- **Word segmentation + TF-IDF + Linear SVM**

Challenge: The positive examples are imbalanced across categories.

TASK 2 — EXPERIMENTAL RESULTS

System	Macro F1
BiLSTM	0.36
Word TF-IDF + Linear SVM	0.4127
Character TF-IDF + Linear SVM	0.6045
Word TF-IDF + Linear SVM	0.5098

Best Reported Model

Character TF-IDF + Linear SVM

TASK 3 — POLARIZATION MANIFESTATION IDENTIFICATION

Question: How is polarization expressed in the text?

Six Target Categories

- **Stereotype**
- **Vilification**
- **Dehumanization**
- **Extreme Language**
- **Lack of Empathy**
- **Invalidation**

Models:

- **Character TF-IDF + One-vs-Rest Linear SVM**
- **Word TF-IDF + One-vs-Rest Linear SVM**
- **FastText binary classifiers for individual labels**

TASK 3 — EXPERIMENTAL RESULTS

System	Macro F1
Character TF-IDF + SVM	0.4323
Word TF-IDF + SVM	0.3474
BiLSTM	0.35

***Challenge**: Identifying subtle or less frequently represented manifestations remains challenging.*

FINE-TUNED MODELS

Test macro-F1 (single seed = 42, 80/10/10 split)

Model	Task-1	Task-2	Task-3
mDeBERTa-v3-base	**0.834**	**0.729**	**0.590**
NLLB-200-600M (encoder)	0.813	0.670	0.570
XLM-R base	0.788	0.720	0.566
mBERT	0.773	0.675	0.579

- mDeBERTa-v3-base is the best model on all three tasks.
- On Task-3 the four models lie between 0.566 and 0.590. With one seed and a small test set, this gap should not be read as a clear ranking.

CROSS-TASK FINDINGS

1. Character-Based Features

Character n-grams performed strongly in Task 2 and Task 3, suggesting their usefulness for Burmese text classification.

2. Neural Models

The Stacked BiLSTM performed best among the listed Task 1 models, but the Task 2 BiLSTM struggled in the reported multi-label experiment.

3. Label Imbalance

- Less frequent categories had weaker predictive performance, highlighting the importance of per-label evaluation.

LIMITATIONS & FUTURE WORK

CURRENT LIMITATIONS

- Inconsistent and malformed source data
- Imbalanced labels
- Limited performance for some minority categories
- Incomplete overall FastText evaluation results in the saved Task 3 notebook
- No reported pretrained transformer fine-tuning results in the summarized experiments

FUTURE WORK

- Audit annotations and resolve disagreements
- Preserve annotator and source metadata
- Explore class weighting and threshold tuning
- Perform detailed per-label error analysis
- Experiment with pretrained multilingual or Burmese language models
- Use consistent evaluation protocols and report complete test metrics

CONCLUSION

This project explored three connected tasks for Burmese polarization analysis using the myPolar dataset.

- Task 1: Stacked BiLSTM achieved a test Macro F1 of **0.7188**.
- Task 2: **Character TF-IDF + Linear SVM** achieved a validation Macro F1 of 0.6252.
- Task 3: Character TF-IDF + Linear SVM achieved a test Macro F1 of 0.4323.

Traditional machine learning with character-based features provided strong baselines, while neural models showed different performance across tasks. Data quality, label imbalance, and careful evaluation remain important challenges.

REFERENCES

- POLAR task definitions: https://polar-semeval.github.io/tasks.html
- Course Assignment 2: https://github.com/ye-kyaw-thu/AIE-F-B2/tree/main/assignment/assignment-2
- Task 1: Task1_PolarizationDetection.ipynb
- Task 2: Task2_TypeClassification.ipynb
- Task 3: Task3_ManifestationIdentification.ipynb

THANK YOU!